

# SafetyVision

A PPE Compliance System for Workplace Safety

TIER 2 | ADVANCED

Hnin Aung | May Pang Kue

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# The Problem

On construction and manufacturing sites, workers are required to wear personal protective equipment (PPE) like hard hats, safety vests, and glasses. Checking compliance is done manually by a safety officer who cannot watch everyone at once, so violations get missed until someone is hurt.

## Who cares?

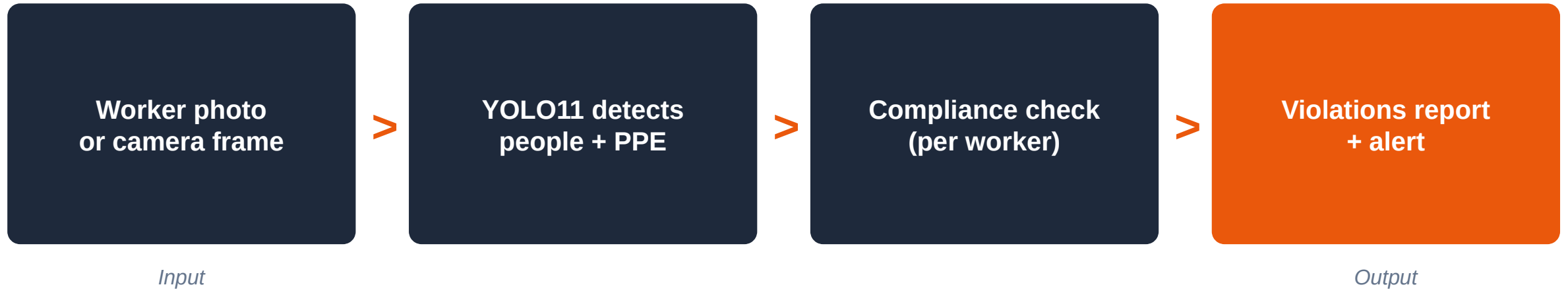
- Safety managers who are responsible for compliance
- Construction and manufacturing companies
- Workers whose safety depends on it

## Why it matters

Head injuries and other PPE-related incidents are among the most common serious workplace accidents, and they lead to lost work time, medical costs, and regulatory fines. A system that automatically flags a missing hard hat or vest from a camera image lets one safety officer cover a whole site and catch problems before they become injuries.

# Our Solution

SafetyVision takes a photo of a work area, detects each worker and their PPE, checks who is missing required gear, and produces a simple violations report.



Input → Model → Application logic → Output

# Technical Approach

## CV Technique

Object Detection (plus a rule-based compliance component)

## Model

YOLO11, fine-tuned on PPE categories via transfer learning

## Framework

PyTorch with the Ultralytics library

## Architecture

Convolutional neural network (CNN)

Why this approach: YOLO11 is the current real-time detection standard, it is free and open source, and transfer learning lets us teach it our PPE categories with a small dataset. The compliance component then turns raw detections into the report a safety officer actually needs, which makes this a Tier 2 two-part system.

# Data Plan

|        |                                                                                      |
|--------|--------------------------------------------------------------------------------------|
| Source | Public PPE datasets on Roboflow Universe, plus a small set of our own labeled photos |
|--------|--------------------------------------------------------------------------------------|

|      |                                                   |
|------|---------------------------------------------------|
| Size | Roughly 3,000 to 6,000 labeled images (free tier) |
|------|---------------------------------------------------|

|        |                                                                                        |
|--------|----------------------------------------------------------------------------------------|
| Labels | Hard hat, safety vest, safety glasses, and person (with 'no-helmet' / 'no-vest' cases) |
|--------|----------------------------------------------------------------------------------------|

|      |                                                           |
|------|-----------------------------------------------------------|
| Link | Roboflow Universe PPE datasets (public, free to download) |
|------|-----------------------------------------------------------|

*We follow the 'demo first' rule: prove the pipeline on a handful of sample images before investing in the full dataset.*

# Success Metrics

## PRIMARY

**mAP@0.5  $\geq$  0.80**

Detection quality: the model should reliably find PPE items. We will measure mean Average Precision at IoU 0.5 and aim for at least 0.80 on our validation set.

## SECONDARY

**Under 1 second / image**

Speed: for the tool to be usable on-site, each image should be processed in under one second on free Colab hardware.

*We will report both numbers honestly, including any classes the model struggles with.*

# Milestone Plan (10-week summer term)

| Phase              | Goal                                               | When      |
|--------------------|----------------------------------------------------|-----------|
| Blueprint          | Plan approved, midterm submitted                   | Week 5    |
| First Working Demo | Pretrained YOLO11 runs on a few sample images      | Week 6    |
| Make It Yours      | Fine-tune on PPE data; add compliance-report logic | Weeks 7-8 |
| Improve & Measure  | Test, fix, and measure against our metrics         | Week 9    |
| Package & Present  | Demo video, README, final slides                   | Week 10   |

*First Working Demo comes before heavy data and training work, on purpose.*

# Risks and Resources

## Risk 1: PPE data is limited or imbalanced

Probability: Medium

Plan B: Combine several Roboflow PPE datasets and use data augmentation (flips, brightness) to expand and balance the classes.

## Risk 2: Model misses small items (glasses)

Probability: Medium

Plan B: Focus the scope on hard hats and vests first (larger, easier), and add glasses only once the core system works reliably.

## Resources

- Compute: Google Colab (free tier, T4 GPU)
- Data & models: Roboflow, Ultralytics YOLO11, all open source
- Estimated cost: \$0 (free tier and open source only)